



Classification of Air Force Aviation Accidents: Mishap Trends and Prevention

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This article reanalyzed 124 U.S. Air Force aviation mishaps from 1992 through 2005, using the Department of Defense Human Factors Analysis and Classification System. Mishap types included controlled flight into terrain, loss of control, spatial disorientation, and midair collisions. All accidents were linked with supervisory errors related to proficiency and experience factors. Preconditions to unsafe acts were most frequently related to cognitive, psychobehavioral, and coordination, communication, and planning factors. Continued improvements in operational risk assessment and crew resource management are suggested, as well as interventions such as flight operations quality assurance and line-oriented safety audits to specifically focus latent error mitigation strategies.

In 2003, the U.S. Secretary of Defense issued a letter titled, “Reducing Preventable Accidents,” which challenged the Department of Defense (DoD) to reduce accident rates by 50% in the next 2 years (Rumsfeld, 2003). General Jumper, the former Chief of Staff of the U.S. Air Force, stated that the Air Force goal is eventually to reach a zero mishap rate: “Any goal other than zero implies that some mishaps are acceptable. But no mishap is” (Jumper, 2004, p. 1). As part of the effort to achieve these ambitious targets, the U.S. Army, Navy, Coast Guard, Marine Corps, and Air Force agreed to adopt a common policy on the collection and analysis of mishap human factors data (Joint Services Safety Chiefs, 2005). This agreement stated that all the services are to incorporate the DoD Human Factors Analysis and Classification System (HFACS) to help mit-

igate human factors hazards. The DoD HFACS framework was adopted to better understand the precursors and underlying causal human factors associated with mishaps.

The objective of this research was to use the new DoD HFACS to determine common precursors or trends in US Air Force accidents. Then use that information to advocate focused improvements in operations that are prevention measures to reduce risk. The concept of going beyond crew actions and inactions is not new. This research is similar to notable studies involving military aviation (Feggetter, 1982; Gerbert & Kemmler, 1986; Holland & Freeman, 1995; Shappell & Wiegmann, 1996; Yacavone, 1993), military and crew resource management (CRM) research (Nullmeyer, Stella, & Montijo, 2005; Wiegmann & Shappell, 1999), research emphasizing prevention measures (Fitts & Jones, 1961; Khatwa & Helmreich, 1999; Kowalsky, Masters, Stone, Babcock, & Rypka, 1974; Moroze & Snow, 1999), and studies specifically addressing HFACS (Shappell & Wiegmann, 2000, 2003; Wiegmann & Shappell, 1997, 2001). This article integrates these ideas to identify and mitigate human causal factors in Air Force operations. We re-analyzed a sample of U.S. Air Force accidents from 1992 to 2005 using the DoD HFACS framework and identified specific latent operational hazards. Prevention measures based on those latent hazards are then proposed with the goal of reducing the mishap rate to save lives, resources, and money.

U.S. AIR FORCE MISHAP INFORMATION

The Air Force defines a Class A mishap as one that involves a fatality or permanent disability, destroys an aircraft, or has a total mishap cost in excess of \$1 million (Air Force Safety Analysis [AFSA], 2003, p. 5). The overall Air Force average Class A mishap rate for the last 10-year period was 1.32 mishaps every 100,000 flying hours (Air Force Safety, 2005). This rate can be considered the probability of a mishap, or the risk. Put another way, it is the time between failures, with one catastrophic failure every 75,757 flying hours. The goal of this study was to contribute to Class A mishap reduction by mitigating risks at all levels, not just focusing on the actions of the pilot(s).

Despite years of CRM and operational risk management (ORM) training and education, causal factors representing those constructs still contribute to mishaps today. In August 2002, an MC-130H special operations aircraft flying a night low-level navigation training mission in a mountainous area of Puerto Rico hit a ridgeline, killing all crewmembers on board (Accident Investigation Board [AIB], 2002). The AIB cited the aircrew's loss of situational awareness and the lack of appropriate response to terrain warnings. Other factors contributing to the mishap were overall crew preparation, misdirected attentional focus on the weather, CRM issues, and poor crew judgment (AIB, 2002). A similar accident occurred in March

2005 when an Air Force MC-130H was flying a night training mission in the mountains of Albania and struck the terrain, resulting in nine fatalities (Rolfesen, 2005b). The AIB concluded that the accident did not occur due to avionics or mechanical problems, but due to “a series of critical errors by the flight deck crew led to the special operations plane being trapped as it flew too low and too slow in the remote valley” (Rolfesen, 2005b, p. 14).

These accidents are examples of controlled flight into terrain (CFIT), defined as an in-flight collision with terrain, water, trees, or man-made obstacles during forward flight (AFSA, 2003, p. 13). Although the aviation community has learned much from previous CFIT accidents (Khatwa & Helmreich, 1999), these mishaps illustrate the sources of human error and aviation organizational failures in current operations. Table 1 depicts the high cost of Air Force CFIT, midair collisions, loss of control, and taxi, take-off, and landing mishaps between 1993 and 2002, in terms of aircraft, personnel, and money.

According to an Air Force Safety Center analysis, from 1993 to 2002, “controlled flight into terrain takes [the] greatest toll: average 6 destroyed aircraft per year and average 13 fatalities per year” (AFSA, 2003, p. 22). CFIT accounts for approximately 22% of all Class A mishaps (AFSA, 2003, p. 11). Note that these categories in Table 1 do not represent underlying causal factors per se, but instead indicate crew activities at the time of the mishap. Just as in civilian aviation, most Air Force accidents are attributed to human error. Unfortunately, previous Air Force safety analysis methods have not been sufficient to identify the underlying causes of the mishaps because they did not sufficiently examine factors beyond crew actions. In fact, one analysis “doubts the Air Force flight-mishap rate can be cut in half because there aren’t widespread common problems” (Rolfesen, 2005a, p. 16). We believe, however, that proper analysis can reveal common problems. Furthermore, it is likely these commonalities represent latent failures that go beyond active errors made by pilots.

TABLE 1
U.S. Air Force 10-Year Totals of CFIT, Midair, Loss of Control, and Taxi,
Take-Off, and Landing Mishaps, 1993–2002

	<i>Destroyed Aircraft</i>	<i>Fatalities</i>	<i>Cost (FY01)</i>
CFIT	59	132	\$1.94 billion
Midair collision	37	90	\$776 million
Loss of control	31	22	\$776 million
Taxi, take-off, and landing	12	5	\$205 million
Total	139	249	\$3.7 billion

Note. CFIT = controlled flight into terrain defined by the Air Force as in-flight collision with terrain, water, trees, or man-made obstacle during forward flight. FY01 = October 1 to September 30. Adapted from Air Force Safety (2003), pp. 13, 15, & 17.)

DEPARTMENT OF DEFENSE HUMAN FACTORS
ANALYSIS AND CLASSIFICATION SYSTEM

DoD HFACS is based on Reason’s (1990) theory of human error and the premise that “simply knowing how past disasters happened does not, of itself, prevent future ones” (p. 17). Reason emphasized the use of case studies married with theory to reduce or eliminate error consequences. The DoD HFACS (2005), shown in Figure 1, is based on the work of Wiegmann and Shappell’s (1997) Failure Analysis Classification System (FACS), which has evolved over the years (1999, 2000, 2001, 2003). For consistency of terms, Figure 1 depicts the HFACS categories, acts, and the subcategories within acts, skill-based error (SB), judgment/decision-making (J/DM) error, perception error, and violation. Not shown in Figure 1 are the specific categories within the subcategories. For instance, for J/DM the specific classification categories are risk assessment, task misprioritization, necessary

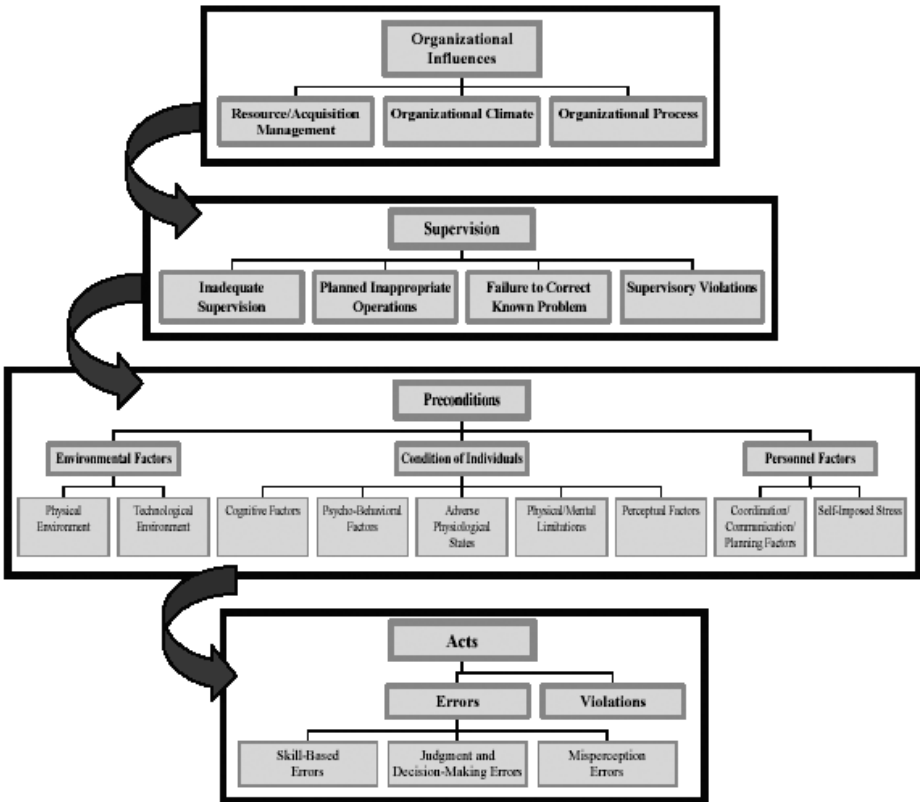


FIGURE 1 DoD HFACS (2005, p. 5)

action-rushed, necessary action-delayed, caution/warning ignored, and decision making during operation. There are a total of 147 specific DoD HFACS categories.

Wiegmann and Shappell's original FACS and the DoD HFACS are similar in theory and basic framework, but the DoD HFACS is more detailed in its classification categories. Prior to adopting DoD HFACS, the U.S. Air Force investigative process focused primarily on active errors made by the pilots and lacked the overall structure that DoD HFACS provides. Active errors are actions or inactions that directly lead to a mishap, whereas latent errors are preexisting conditions that may set up a pilot to fail through a series of events. The previous Air Force safety investigative system was useful and led to many safety improvements, but DoD HFACS is an improvement because it "bridges the gap between theory and practice" (Shappell & Wiegmann, 2000, p. 13). Shappell and Wiegmann (2000) argued that for a safety program to improve the mishap rate, investigations must explore all levels of the accident chain (supervision and organizational) and not simply focus on the final actions of the pilot(s).

DoD HFACS captures the essence of the "domino theory" described by Brown (1995) that depicts accidents as part of a chain-of-events schema and stresses the combination of people and objects within an environment, as well as the temporal nature of their interaction. The domino theory has been modernized and incorporated in DoD HFACS as a "Swiss cheese" model of human error based on Reason's (1990) model shown in Figure 2 (DoD HFACS, 2005). The DoD HFACS model

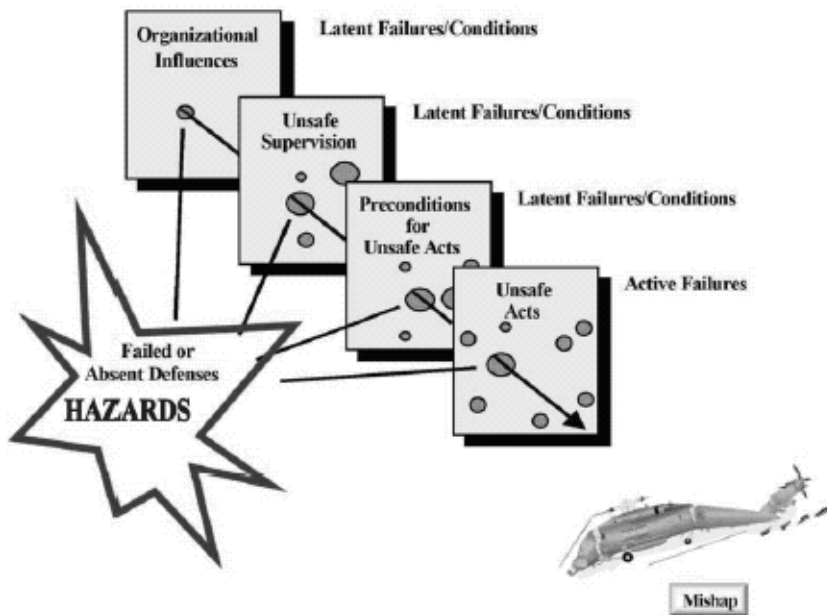


FIGURE 2 Swiss cheese model (DoD HFACS, 2005, p. 4).

identifies specific deficiencies (latent failures) that lead to the “holes in the cheese,” and when combined with certain events, align to produce incidents or accidents. The primary goal of this research was to use DoD HFACS as a framework to identify common latent errors for the purpose of mishap prevention.

METHOD

This study examined a sample of 124 U.S. Air Force Safety Class A mishaps occurring between October 1992 and March 2005. Data reviewed included accident reports as well as summaries and findings of causes prepared by the Air Force Safety Investigation Board (SIB). These reports were retrieved from the U.S. Air Force Safety Center's secure accident database and reanalyzed to identify causal findings using DoD HFACS. Only those 124 Class A mishaps that fell into the categories of CFIT, spatial disorientation (SD), loss of control (LoC), or midair collisions were sampled. One mishap extensively documented by Kern (1998) was not found in the Air Force data locations but was included in the study. This was the well-publicized April 1996 CT-43 crash in Dubrovnik, Croatia, in which 6 Air Force crewmembers and 29 passengers were killed. Kern's investigation was used to code this CFIT mishap into DoD HFACS.

These accidents were not reinvestigated, only recoded based on a summary of the accident and a list of findings and causal factors. This summary provided the SIB's perspective of the most important findings and causes and a macroassessment of each mishap. In addition to the computer database of the accident summaries, a review of the actual SIB reports kept at the U.S. Air Force Safety Center, Kirtland Air Force Base, New Mexico, was also completed. It is emphasized that the previous Air Force accident classification system, although lacking in the structure provided by DoD HFACS, did provide many of the same human factors concepts and terminology. Consequently, the recoding was a straightforward process with little deliberation because DoD HFACS contains very specific descriptions and definitions of each category.

The first author accomplished the recoding into DoD HFACS and served as the subject matter expert. His experience as a 20-year Air Force pilot in five different aircraft, an academic background in human factors, and attendance at a 1997 FACS workshop allowed for reliable classification. The process consisted of examining each mishap and evaluating its recoding into one of the 147 different DoD HFACS categories. Based on the actual Air Force accident report, the information was recoded with a 0 (*no*) or 1 (*yes*) as the binary process of converting previous mishap information into DoD HFACS. Subcategories were marked several times for a mishap; however, specific categories were only marked once per mishap. The mishap itself, the complexity of the causal chain of events, and the manner in which the investigation was written all influenced the total number of classifica-

tions. Some mishaps had relatively few marked categories and others had multiple DoD HFACS classifications.

Mishap Types

For the purpose of this study, CFIT is broken into two different categories, one during enroute flight and one an approach and landing accident (ALA). We defined CFIT-1 as previously presented, which is comparable to the Air Force definition. We defined CFIT-2/ALA as an airworthy aircraft unintentionally colliding with terrain, water, trees, or a man-made obstacle during controlled flight in the approach and landing phase of flight. This distinction was made to cover the Air Force category of approach and landing accidents with a similar definition that is used in the commercial accident framework (Khatwa & Helmreich, 1999). Other mishap types recorded were LoC due to stall, spin, or any other out-of-control condition; SD, which also includes g-induced loss of consciousness; and midair collisions.

RESULTS

The classification of the active and latent errors of the accidents produced frequencies that were analyzed by mishap type. Of the 124 mishaps, 109 (88%) were training missions and the remainder were operational or combat losses. The 109 training mission incidents resulted in 173 fatalities. Table 2 summarizes the aircraft by mishap type. Combined CFIT accidents (CFIT-1 & CFIT-2/ALA) occurred most often, with 48 of the total 124 mishaps attributed to CFIT. Midair collisions accounted for 42 mishaps out of 124, followed by CFIT-1 with 31, SD with 19, CFIT-2/ALA with 17, and LoC with 15. Fighter, attack, and reconnaissance (FAR) aircraft had by far the highest frequency of midair collisions and SD. Crew aircraft

TABLE 2
Total of Each Mishap Type by Aircraft Category

	<i>CFIT-1</i>	<i>CFIT-2/ALA</i>	<i>LoC</i>	<i>SD</i>	<i>Midair</i>	<i>Total</i>
FAR	17	6	9	19	34	85
Crew	8	6	0	0	2	16
Helicopter	6	3	1	0	2	12
Trainer	0	2	5	0	4	11
Total	31	17	15	19	42	124

Note. CFIT = controlled flight into terrain; ALA = approach and landing accident; LoC = loss of control; SD = spatial disorientation; FAR = fighter, attack, and reconnaissance aircraft; crew = multiengine, heavy aircraft.

had 16 Class A mishaps, with 14 of them CFIT, and 8 of those were CFIT-1. Trainer aircraft exhibited a pattern similar to FAR aircraft, with midair collision and LoC occurring more frequently.

DoD HFACS

The following sections provide the frequency data for factors related to unsafe acts, preconditions to unsafe acts, and the associated supervisory and organizational influences. The causal factors from the 124 mishaps were classified into 102 of the possible 147 specific DoD HFACS categories.

Acts. DoD HFACS defines acts as active failures or actions committed by the operator that result in human error or an unsafe situation (DoD HFACS, 2005, p. 6). Table 3 depicts the number of mishaps within each mishap type that had at least one causal factor associated with each subcategory. This method of depicting relative frequencies was used because a subcategory could have multiple classifications for the same mishap. For CFIT-1, J/DM errors were found in 26 of the 31 mishaps and J/DM errors also led the CFIT-2/ALA mishap type, found in 14 of the 17 mishaps. LoC and midair collisions were most often associated with SB errors. SB errors were found in 12 of the 15 LoC mishaps and 40 of the 42 midair collision accidents. Note that violations occurred more frequently in CFIT-2/ALA and LoC mishaps compared to other mishap types.

Table 4 presents a further breakdown of error classification for the major subcategories within each type of unsafe act. Note that the number depicted by each act represents the number of total causal factors for each Act (specific categories were only coded once per mishap). The percentages listed next to each specific category reflect the percentage relative to this total number of causal factors. Recall J/DM errors were the most prevalent causal factor in CFIT-1 and CFIT-2/ALA. Within that error subcategory, Table 4 depicts risk and decision-making errors as

TABLE 3
DoD HFACS Category Classification of Unsafe Acts by Mishap Type
(Number of Mishaps With Subcategory Instances for the Mishap Type)

	<i>CFIT-1</i>	<i>CFIT-2/ALA</i>	<i>LoC</i>	<i>SD</i>	<i>Midair</i>
No. of mishaps	31	17	15	19	42
Skill-based	20	9	12	9	40
J/DM	26	14	8	11	24
Perception	11	9	3	16	17
Violation	5	7	6	2	3

Note. CFIT = controlled flight into terrain; ALA = approach and landing accident; LoC = loss of control; SD = spatial disorientation; J/DM= judgment/decision making.

TABLE 4
DoD HFACS Classification by Number of Instances Within Specific
Category (Number of Coded Specific Categories/Number of Total
Instances in Subcategory)

	<i>CFIT-1</i>	<i>CFIT-2/ALA</i>	<i>LoC</i>	<i>SD</i>	<i>Midair</i>
Skill-based	<u>23</u> Proc 39% Scan 39%	<u>14</u> Proc 50% Ov/Un 29%	<u>16</u> Proc 44% Ov/Un 38%	<u>10</u> G-strn 50% Scan 20%	<u>44</u> Scan 82% Proc 18%
J/DM	<u>45</u> Risk 38% DM 31% TskMP 22%	<u>24</u> Risk 46% DM 33%	<u>6</u> DM 50% Risk 23%	<u>16</u> TskMP 56% Risk 19%	<u>28</u> TskMP 56% DM 21%

Note. CFIT = controlled flight into terrain; ALA = approach and landing accident; LoC = loss of control; SD = spatial disorientation; J/DM = judgment/decision making; Proc = procedure; Scan = breakdown of visual scan; Ov/Un = over/under control; G-strn = improper anti-G straining technique; Risk = risk assessment; DM = decision making; TskMP = task misprioritization.

the leading specific classifications contributing to mishaps. LoC was led by SB factors with procedural and over/under control errors as the most frequent SB error acts. SD mishaps were lead by perception errors followed by J/DM errors of task misprioritization. Finally, midair collisions were most frequently classified by the SB error of breakdown in visual scan as well as the J/DM error of task misprioritization.

Preconditions to unsafe acts. Table 5 shows the preconditions to unsafe acts, defined as factors in a mishap if conditions of operators, environment, or per-

TABLE 5
DoD HFACS Category Classification of Preconditions to Unsafe Acts
by Mishap Type (Number of Mishaps with Subcategory Instances
for the Mishap Type)

	<i>CFIT-1</i>	<i>CFIT-2/ALA</i>	<i>LoC</i>	<i>SD</i>	<i>Midair</i>
No. of mishaps	31	17	15	19	42
Cognitive	22	13	5	15	31
Psycho-behav	19	11	6	6	14
Perceptual	20	12	6	15	7
CCP	20	12	4	3	17
Phys Envi	13	12	1	12	10

Note. Only most frequent five of nine factor areas shown. CFIT = controlled flight into terrain; ALA = approach and landing accident; LoC = loss of control; SD = spatial disorientation; Psycho-Behav = psychological-behavioral factors; CCP = coordination, communication, and planning factors; Phys Envi = physical environmental factors.

sonnel factors affect practices, conditions, or actions and result in human error or an unsafe situation (DoD HFACS, 2005, p. 7). Precondition latent errors had by far the most possible classification categories with 92 of the 147 total. Table 5 depicts a fairly even spread for CFIT-1, CFIT-2/ALA, and LoC mishaps, with cognitive, psychobehavioral, perceptual, and coordination, communication, and planning (CCP) factors all having similar number of mishap classifications relative to the total within each mishap type. SD mishaps were more concentrated in cognitive and perceptual subcategories and midair collisions were heavily frequented by cognitive latent factors.

Table 6 presents the breakdown of subcategories into the more informative specific DoD HFACS classifications of latent error contributions. These specific categories could only be marked once per mishap, therefore the percentages are based

TABLE 6
DoD HFACS Classification by Number of Instances Within Specific
Category (Number of Coded Specific Categories/Number of Total
Instances in Subcategory

	<i>CFIT-1</i>	<i>CFIT-2/ALA</i>	<i>LoC</i>	<i>SD</i>	<i>Midair</i>
Cognitive	<u>38</u> ChnAt 48% Dstrct 21% Inatt 18%	<u>23</u> ChnAt 48% TskOv 17% Dstrct 17%	<u>7</u> ChnAt 57%	<u>22</u> ChnAt 55% TskOv 23% Dstrct 18%	<u>51</u> ChnAt 47% Inatt 22% TskOv 16%
Psychobehavioral	<u>29</u> Compl 24% Oconf 21% RspSt 21%	<u>21</u> ExSucc 24% Compl 19% Press 15%	<u>15</u> Oconf 27% Compl 27% Oaggrs 27%	<u>5</u> ExSucc 40% Compl 18%	<u>16</u> RspSt 50%
Perceptual	<u>31</u> MspCnd 61% SDO 19%	<u>21</u> MspCd 62% VsIll 24%	<u>6</u> MspCd 50% SDO 50%	<u>30</u> SD 50% MispCnd 27%	<u>7</u> MspCd 43%
CCP	<u>46</u> MsnPln 24% Brf 15% Repln 13% Ldshp 13%	<u>33</u> Monitr 21% Plan 18% Repln 12% Assert 12% Ldshp 12%	<u>7</u> Ldshp 29% Monitr 29%	—	<u>23</u> Miscm 30% Ldshp 26% Plan 13%
Phys Envi	<u>13</u> MC 100%	<u>12</u> MC 100%	<u>1</u> MC 100%	<u>12</u> MC 100%	<u>12</u> MC 83%

Note. CFIT = controlled flight into terrain; ALA = approach and landing accident; LoC = loss of control; SD = spatial disorientation; Assert = assertiveness; Brf = briefing; CCP = coordination, communication, and planning factors; ChnAt = channelized attention; Compl = complacency; Dstrct = distraction; ExSucc = excessive motivation to succeed; Inatt = inattention; Ldshp = leadership; MC = meteorological conditions; Miscm = miscommunication; MspCd = misperceived conditions; Monitr = monitor other pilot; MsnPln = mission planning; Oaggrs = over aggressive; Oconf = over confidence; Phys Envi = physical environment; Press = pressing; Repln = replan mission during mission; RspSt = response set; TskOv = task oversaturation; VsIll = visual illusion.

on the total number of coded causal factors. For CFIT-1 mishaps, Table 5 depicted cognitive factors occurred in 22 of the 31 mishaps and CCP factors were present in 20 of 31 mishaps; however, Table 6 shows that of the causal factors coded, the CCP factor had the most instances. Within the CCP factor, mission planning was the leading latent error contributor. Note the frequent appearance of leadership as a latent CCP precondition in the mishap chain for four of the five mishap types.

CFIT-2/ALA followed a similar pattern to CFIT-1; however, within CCP the most frequently classified latent error was monitoring, observing, and challenging the performance of another pilot. Also for CFIT-2/ALA, 12 of the 17 mishaps had the physical environment latent factor of vision restricted by meteorological conditions, defined as restricted due to weather, haze, or darkness (DoD HFACS, 2005, p. 1-5).

LoC was led by the psychobehavioral factor, which is defined as an individual's personality traits, psychosocial problems, psychological disorders, or inappropriate motivation creating an unsafe situation (DoD HFACS, 2005, p. 1-8). Within the psychobehavioral category for LoC were specific classifications of overconfidence, overaggressiveness, and complacency.

Cognitive factors were frequently associated with all accident types but were most often associated with midair collisions, as shown in Table 6. Within cognitive factors the most frequent latent error was channelized attention, the most frequent cognitive classification of all five mishap types. Channelized attention is defined as an individual focusing all conscious attention on a limited number of environmental cues to the exclusion of others of a subjectively equal or higher or more immediate priority, leading to an unsafe situation (DoD HFACS, p. 1-7).

Supervisory and Organizational influences. Supervisory influences are defined as methods, decisions, or policies of the supervisory chain-of-command directly affecting practices, conditions, and actions resulting in human error or an unsafe situation (DoD HFACS, 2005, p. 1-18). This category of DoD HFACS directly contributes to preconditions of unsafe acts discussed in the previous section and thus is critical to aviation safety. Unfortunately, the previous U.S. Air Force Safety investigation system did not consistently document findings in this area. Therefore, analysis of supervisor as well as organizational influences was difficult and somewhat limited.

Of the four subcategories shown in Figure 1, most fell into the categories of inappropriate operations and inadequate supervision shown in Table 7. For all mishaps, inappropriate operations was the most frequently classified factor.

Table 8 further breaks down the classification of specific categories within the subcategory factors. The most common latent contributing errors by supervisors were mission risk assessment, pilot limited recent and total experience, and pilot proficiency. Inadequate supervision specific classifications were local training issues/programs and supervision policies.

TABLE 7
DoD HFACS Category Classification of Preconditions to Unsafe Acts
by Mishap Type (Number of Mishaps With Subcategory Instances
for the Mishap Type

	<i>CFIT-1</i>	<i>CFIT-2/ALA</i>	<i>LoC</i>	<i>SD</i>	<i>Midair</i>
No. of mishaps	31	17	15	19	42
Inappropriate operations	14	10	7	8	10
Inadequate supervision	6	4	6	4	6

Note. CFIT = controlled flight into terrain; ALA = approach and landing accident; LoC = loss of control; SD = spatial disorientation.

TABLE 8
DoD HFACS Classification by Number of Instances Within Specific
Category (Number of Coded Specific Categories/Number of Total
Instances in Subcategory

	<i>CFIT-1</i>	<i>CFIT-2/ALA</i>	<i>LoC</i>	<i>SD</i>	<i>Midair</i>
Inappropriate operations	<u>32</u> Risk 31% Prof 25% Exp 19%	<u>15</u> Prof 40% Exp 40%	<u>14</u> Prof 50% Risk 21% Exp 21%	<u>14</u> Exp 43% Prof 36%	<u>17</u> Prof 35% Exp 35%
Inadequate supervision	<u>6</u> InSup 83% LTrn 17%	<u>6</u> InSup 66% LTrn 17% Feedbck 17%	<u>8</u> InSup 50% LTrn 25%	<u>4</u> LTrn 50% Pol 25%	<u>5</u> LTrn 40% Pol 40%

Note. CFIT = controlled flight into terrain; ALA = approach and landing accident; LoC = loss of control; SD = spatial disorientation; Exp = total and recent experience; Feedbck = feedback; InSup = inadequate supervision; LTrn = local training; Pol = policy; Prof = proficiency; Risk = risk assessment.

Organizational influences are not specifically discussed due to the difficulties encountered in reclassification of the previous Air Force system into DoD HFACS at the organizational resource, process, and climate levels. Although, there were thorough discussions regarding organizational influences for some mishaps in the previous Air Force system, their scope was limited to allow credible classification into the DoD HFACS framework. In general, the limited analysis indicated potential deficiencies in the categories of publications and training as well as resource acquisition of fleetwide modifications and training devices.

DISCUSSION

Reason (1990) described latent errors as “the significance of causal factors present in the system before an accident sequence actually begins” (p. 197). Reclassifica-

tion of previous USAF accidents within the DoD HFACS framework provides the data to target root causes and improve mishap prevention programs. To make the best use of available resources, prevention measures should focus on the areas with the greatest return, the holes in the cheese that are most manageable, and those where the precursors are more susceptible to an antidote. DoD HFACS (2005) describes the trends of contributing factors in accidents: "If you know what these system failures/hazards or 'holes' are, you can better identify their roles in mishaps—or better yet, detect their presence and develop risk mitigation strategy correcting them *before* a mishap occurs" (p. 4).

Although studies of active errors have value (Fitts & Jones, 1961; Khatwa & Helmreich, 1999; Kowalsky et al., 1974; National Transportation Safety Board [NTSB], 1994; Zeller, 1981), errors are best prevented by addressing the organizational, supervisory, and preconditions that lead to these unsafe acts. Technology can reduce error; however, it often merely changes the nature of errors and allows for expression of new active error types. Reason (1990) and the research model of Shappell and Wiegmann (2000) stress elimination of the organizational latent hazards to mitigate risk, or elimination of the holes in the cheese further up the chain of events. This reanalysis of U.S. Air Force mishap data has allowed us to identify the latent factors that contribute to observed mishap types. Table 9 summarizes the mishap types with the most frequent HFACS error categories and best represents the alignment of the holes in the cheese, the latent-to-active error sequence that shaped the 124 mishaps assessed in this study.

CFIT-1 and CFIT-2/ALA

Our analysis suggests that CFIT accidents are most often associated with unsafe acts in the category of J/DM. The main Preconditions to these unsafe acts were CCP factors, all of which are Air Force CRM topic areas. This prevalence is somewhat surprising as CRM training has been emphasized in all aviation communities over the last 20 years. The Air Force guidance on CRM describes the six dimensions of CRM as situational awareness, crew coordination/flight integrity, communication, risk management/decision making, task management, and mission planning/debriefing (Air Force Instruction [AFI] 11–290, 2001, p. 5). Yet, those areas frequently contributed to the mishap types analyzed in this study, particularly recent CFIT mishaps.

Our findings parallel Nullmeyer et al.'s (2005) work examining CRM in fighter (F-16 and A-10) and transport (C-130) aircraft. In their assessment of 21 F-16 mishaps and 19 A-10 mishaps, they found situational awareness, task management, and risk management/decision-making CRM dimensions most frequent. They concluded that CRM-related human factors are areas to focus on in mishap prevention measures.

TABLE 9
Summary of Most Frequent HFACS Categories Contributing
to a Mishap Sequence

	<i>HFACS Category</i>		
	<i>Supervision (Latent)</i>	<i>Precondition (Latent)</i>	<i>Act</i>
CFIT-1	Risk Proficiency	CCP: ^a Mission Planning	J/DM: Risk DM
CFIT-2/ALA	Proficiency Experience	CCP: ^a Monitoring	J/DM: Risk DM
LoC	Proficiency	Psychobehavioral ^a Over-Confid/Aggress Complacency	SB: Procedure Ov/Un Cont
SD	Experience Proficiency	Perceptual ^a SD	J/DM: Task MP Perception
Midair	Proficiency Experience	Cognitive: ChnAt	SB: Visual scan

Note. CFIT = controlled flight into terrain; ALA = approach and landing accident; LoC = loss of control; SD = spatial disorientation; CCP = coordination, communication, and planning factors; ChnAt = channelized attention; J/DM = judgment/decision making; Over-Confid/Aggress = overconfidence and overaggressive; Ov/Un Cont = over/under control; SB = skill-based; Task MP = task misprioritization.

Cognitive factors associated with CFIT can be addressed by drawing on CFIT reduction training developed in civilian aviation. Khatwa and Helmreich (1999) accomplished analysis of 287 fatal ALAs from 1980 to 1996, similar to our CFIT-2/ALA category, as well as 76 ALAs and serious incidents from 1984 to 1997. Their report concluded with recommendations for aircraft equipment, instrument approach plates and approach procedure changes, stabilized approach criteria, and a pilot education program (pp. 50–51). The training program was delivered throughout the aviation industry with the goal of reducing ALA/CFIT by 50% in 5 years (“FSF task force recommends,” 1998, p. 5).

Technology provides assistance with attention, task management, and perceptual factors by identifying unsafe conditions, depicting terrain and obstacles, and alerting pilots to unsafe parameters. Following the CT-43 crash in Croatia, the Air Force embarked on a program to equip all passenger and troop-carrying aircraft with ground proximity warning systems (personal communication, USAF/XO Memo, September 9, 1996). Most of these aircraft are or will be equipped with the newer generation terrain awareness and warning systems (TAWS), which provide a visual depiction of terrain and a “look ahead” capability.

Head-up displays (HUDs) also mitigate attentional and perceptual issues associated with CFIT as they can bring attentional focus off the instrument panel and allow focus on the external environment. The HUD can project a visual approach-landing glide path, which can be of great help when flying a nonprecision approach or a visual approach in less than optimal visual conditions. Khatwa and Helmreich (1999) found that 75% of CFIT accidents occurred when flying a visual or nonprecision approach. HUDs are common in FAR aircraft and they are also beginning to be installed in some commercial transport aircraft as well.

Procedural mitigation strategies can also be applied. For example, most commercial airlines have established standards, stabilized approach criteria that address required airspeed, glide path, altitude, and vertical velocity “windows” to continue an approach or accomplish a missed approach (Khatwa & Helmreich, 1999). The Air Force has performance skill requirements for evaluation purposes, but it does not currently have formal guidance on implementation of stabilized approach criteria and procedures for use as decision-making aids.

Another aircrew operational policy change to reduce mishap risk is adopting procedures that require the copilot or first officer (the less senior pilot) to fly during challenging portions of the mission. This would allow the senior pilot to accomplish risk assessment and handle tactical decision-making issues that are commonly cited as causal in mishaps. J/DM errors were more prevalent in CFIT-1, CFIT-2/ALA, and SD mishaps compared with SB errors. This is highlighted by the fact that leadership and pilot monitoring were frequently cited in CFIT mishaps. Contributing to the mishap chain is not a lack of skills but rather a lack of risk assessment, improper decisions, and nonassertive communication from other pilots who are not flying.

According to an NTSB (1994) report of flight-crew-involved major accidents from 1978 to 1990, the captain or senior pilot was flying in 30 of 37 accidents (81%; p. 38). The NTSB report and Berman (1995) both concluded that a senior pilot would be a more proactive communicator to monitor and challenge a junior pilot during an approach and landing. Swauger (2003) suggested pilots fail to recognize their poor decisions and incorrectly apply common heuristics to unique situations, providing more evidence that a senior nonflying pilot would be better able to monitor and make prudent, assertive decisions.

Midair Collisions

SB errors, specifically breakdown in visual scanning procedures, were the unsafe act most commonly associated with midair collisions. Another major type of unsafe act was the J/DM factor of task misprioritization. The associated preconditions were the cognitive factors of channelized attention, inattention, and task oversaturation. Another precondition to unsafe act was the role of CCP categories of communication, leadership, and planning.

The primary supervisory influences were inappropriate planned operations as expressed by the subfactors of a lack of proficiency and experience. Given that the Air Force will continue to use relatively young and inexperienced pilots, training should focus on task management. An additional area of interest was the presence of inadequate supervision in the specific category of training and policies. Development of improved procedures and training may be an effective means of mitigating areas such as a lack of experience and proficiency.

Technology can also overcome some of these factors. Studies have shown that the human ability to detect and avoid potential collisions may be poor, especially in complex and fast-paced fighter operations (Morris, 2005). Although most Air Force transport and trainer aircraft are equipped with TCAS to counter these known deficiencies, fighter aircraft do not have a similar system.

Spatial Disorientation

Perception and judgment errors were the unsafe acts most frequently identified in SD mishaps. The preconditions for these acts included cognitive and perceptual factors related to attention and misperception. Once again, limited proficiency and lack of experience were the most common supervisory factors related to these preconditions. In other words, lack of proficiency and relative experience put Air Force pilots at risk for attentional and perceptual problems that lead to judgment errors and misperceptions.

Providing training to help aircrew identify and mitigate SD risk factors continues to be an important Air Force program. Much of this training is provided by U.S. Air Force aerospace physiologists on a recurrent basis. For many years the Air Force aerospace physiology training was required once every 3 years. This training consisted of an academic portion with a review of pilot "airworthiness" issues ranging from human performance, vision, mental or physical fatigue, visual or vestibular misperceptions and night flying (AFI 11-403, 2001). This training was then followed by a "flight" in an altitude chamber to review symptoms of hypoxia and an actual rapid-decompression scenario. This training requirement was extended to once every 5 years (AFI 11-403, 2001).

Beyond SD-specific accidents, SD was also a contributing factor to CFIT-1, CFIT-2/ALA, and LoC mishaps. Numerous safety boards recommended improved SD training devices. Aerospace physiology training is conducted by human factors specialists equipped to address topics commonly listed as preconditions to unsafe acts. Hypoxia is a rare event and the rationale to extend training for it as an environmental hazard is logical; however, the associated academic SD training that accompanies it is vital education for pilots and should be provided on a more regular basis. In addition to training, continued efforts must be made in attentional cuing and other measures to counter SD.

Matthews, Previc, and Bunting (2002) presented results from a U.S. Air Force SD survey they administered to 2,582 pilots. Their survey found the most frequent visual illusions were sloping horizon, atmospheric blending, and black hole (featureless terrain) approaches. The most frequent nonvisual SD were the leans, Coriolis illusion, and G-excess illusion. Matthews et al. concluded that different types of aircraft had different types of SD, and training programs must address the unique aviation operations. Gillingham (1992) also advocated improved training for SD due to increased night operations and the role SD plays in aviation mishaps. This assessment is similar to CRM recommendations regarding more specifically tailored training per aircraft type (Nullmeyer et al., 2005; Wiegmann & Shappell, 1999). Gibb (2007) and Gibb Schvaneveldt, and Gray (2008) recommended visual SD training occur in flight simulators to demonstrate to pilots their perceptual limitations during impoverished visual conditions.

Operational Risk Management

The role of supervisors in mishap prevention is an area of great concern. This study indicated that supervisors contributed to mishaps by failing to adequately mitigate limitations on aircrew proficiency and experience. Additionally, inadequate supervision, training, and policies contributed to all accident categories. Due to limitations in the existing data we were not able to draw strong conclusions regarding common trends in supervisory and organizational factors. It is vitally important that future safety investigations are trained for, and actively assess supervisory and organizational factors. Continuing and improving on ORM assessments at all levels of the organization is a critical preemptive measure to counter these latent errors because ORM allows flying unit leadership to continuously assess their organization on a daily basis for safe and prudent operations (Gibb, 2006).

Data Collection Before the Accident Occurs

The focus of this research was on latent errors. This process is reactive. A medical analogy has been used to describe mishap analysis as similar to performing an autopsy; accident classification is postmortem in that anything learned is too late for that particular patient (International Civil Aviation Organization [ICAO], 2002). Also improvements made after an accident fix the “last accident” but may fail to address other system failures. Describing mishaps is pointless unless proactive measures are also taken to improve the aviation system. Incident analysis is more effective than mishap analysis because it highlights system limitations and weaknesses prior to a mishap occurring (ICAO, 2002). Two growing programs in aviation that are *proactive* are flight operational quality assurance (FOQA) and line operational safety audits (LOSA). These two safety initiatives collect data during normal operations prior to incidents and accidents and allow for the measurement

and analysis of relevant trend information. The data translate into information of potential hazards and risks resulting in proactive mitigation measures. The Air Force does not currently have programs equivalent to FOQA and LOSA.

FOQA programs obtain and analyze data recorded during flight operations to help determine how the aircraft is being operated in terms of the pilot's actions, the aircraft's responses, and the environment within which the aircraft operates (Flight Safety Foundation [FSF], 2004). In terms of the medical analogy, FOQA is a continuous diagnostic assessment of a healthy system. FOQA tracks pilot performance for procedure compliance as well as environment operating situations to aid in training. FOQA records aircraft system component performance and communications with air traffic controllers (ATCs). FOQA has been used to identify unstabilized approaches, flap overspeeds, excessive banking, engine overtemperatures, TAWS alerts, and glide path deviations (FSF, 2004).

FOQA is not intended to "blame" pilots. The data are analyzed and assessed for the purpose of trend identification and safety improvement. Examples of proactive measures taken to improve operations based on FOQA data had one airline determining that more procedural noncompliance was occurring during visual approaches compared to instrument approaches and hence training was refocused (U.S. General Accounting Office [US GAO], 1998). Another airline found that at a certain airport the required descent rate was being exceeded and on further investigation it was found that ATC were to blame due to their handling of inbound aircraft (U.S. GAO, 2004).

LOSA is similar to an annual flight physical, periodically using expert observers to collect data on flight crew performance as the pilots interact with the aircraft, the operational environment, and each other (Federal Aviation Administration [FAA], 2006). LOSA is not a check-ride, only an observation of how operations are being conducted in terms of CRM concepts analyzed with a tool based on threat and error management (ICAO, 2002). CRM is in place to deal with recognized error occurrences (FSF, 2005) and LOSA is a process to capture how successful operations are accomplished rather than the negative or reactive approach of accident investigation. LOSA can help an aviation operation by identifying threats to the operating environment, operating procedures, assessing transfer of training to operations, human-machine interface, safety margins, and pilot work-arounds (FAA, 2006, pp. 3-4). The addition of preaccident/incident data collection programs would further help identify and mitigate specific trends in unsafe acts or precursors at all levels.

CONCLUSION

Implementation of the DoD HFACS analysis framework provides a clear understanding of the root causes at various levels from unsafe acts through organizational influences. It can also be used to propose associated mitigation strategies.

This research reanalyzed existing U.S. Air Force mishap data using the DoD HFACS framework and proposed focused solutions to common causal latent factors. The FY 2005 mishap rate of 1.44 (Rolfen, 2005a) was higher than the last 10-year average of 1.32. Aviation safety concerns are not limited to only the Air Force. In 2006, the U.S. Navy ordered a mandatory “Navy-wide safety stand-down” in response to six crashes resulting in seven fatalities within the first 2 months of the year (McMichael, 2006, p. 32).

Dekker (2003) discounted error classification and rightfully so if it stops at only describing past failures and blaming aviators involved. This research, accomplished by Air Force pilots, attempts to link descriptive safety reports to existing organizational latent factors. Dekker (2005) investigated in detail the Alaskan Airlines Flight 261 accident in 2000. He used this mishap to example an organization’s drift into failure. Our assessment of 124 mishaps was a macroassessment of how current initiatives (e.g., CRM/ORM), in place to improve aviation operations, are not stopping a possible drift toward failure. This research never expected to expose CRM and ORM failures, but illuminating their limitations may encourage implementation of needed proactive FOQA and LOSA safety programs for the collection of data during normal operations prior to incidents and accidents.

Reason (1990) stated, “disasters are very rarely the product of a single monumental blunder” (p. 17); thus no single act can expect to improve aviation safety. Therefore, a systemic approach to aviation safety is needed at all levels, organizational and supervisory, as well as among individual pilots, to reduce the mishap rate. No accident is acceptable and improvements in safety programs and updating aircraft with currently available technology can eliminate the holes in the cheese and reduce the risk of a mishap.

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